

Preparing the Analytic Sample

Replication File I for the TEPS School Distinction Paper

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1 Setup

```
## Load packages
library(tidyverse)
library(haven)

## Store root (override with an environment variable if the store moves)
raw_root <- Sys.getenv("RAWDATA_ROOT", path.expand("~/Analyses/RawData"))

## file_stamp(): provenance stamp for input files
file_stamp <- function(...) {
  paths <- c(...)
  fi <- file.info(paths)
  data.frame(
    file      = basename(paths),
    size_mb   = round(fi$size / 1024^2, 1),
    modified  = format(fi$mtime, "%Y-%m-%d"),
    md5       = unname(tools::md5sum(paths)),
    row.names = NULL
  )
}

## pin_dta_timestamp(): fix the save time in a .dta header so its md5 depends only on the da
pin_dta_timestamp <- function(path, timestamp = "01 Jan 2000 00:00") {
  bytes <- readBin(path, "raw", file.size(path))
  pos <- grepRaw("<timestamp>\021", bytes[1:200], fixed = TRUE) # tag + length byte (17)
  stopifnot(length(pos) == 1, nchar(timestamp) == 17)
  bytes[(pos + 12):(pos + 28)] <- charToRaw(timestamp)
  writeBin(bytes, path)
}
```

2 Student Survey Data

```
## Stamp student data
path_stu <- file.path(raw_root, "TEPS", "w1_j_s_lv7.0.dta")
file_stamp(path_stu)
```

	file	size_mb	modified	md5
1	w1_j_s_lv7.0.dta	7.9	2021-07-12	b04edf69c84965fe258f1f0e66263937

```
## Read student data
wls <- read_dta(path_stu) # wave 1 (2001), n = 20055
#sapply(wls, function(y) sum(is.na(y))) %>% data.frame(na_count = .) %>% filter(na_count != 0)

## Gender (male=1)
wls$male <- case_match(wls$wls502, 1 ~ 1, 2 ~ 0, 99 ~ NA)
```

Warning: `case\_match()` was deprecated in dplyr 1.2.0.
i Please use `recode\_values()` instead.

```
#wls %>% count(male)

## Siblings
for (i in 1:4) {
  wls[[c('edbro', 'ygbro', 'edsis', 'ygsis')[i]]] <- case_match(
    wls[[c('wls203', 'wls204', 'wls205', 'wls206')[i]]],
    1 ~ 0, c(2, 3, 4, 5) ~ 1, c(97, 99) ~ -5)
  #wls %>% count(edbro)

  wls$sib <- wls$edbro + wls$ygbro + wls$edsis + wls$ygsis # all NA = -20
  wls$onlychild <- case_match(wls$sib, 0 ~ 1, -20 ~ NA, .default = 0)
  #wls %>% count(onlychild)

  wls$edsib <- wls$edbro + wls$edsis - wls$onlychild # those with no elder siblings and not t
  wls$edstchild <- case_match(wls$edsib, 0 ~ 1, NA ~ NA, .default = 0)
  #wls %>% count(edstchild)

  wls$ygsib <- wls$ygbro + wls$ygsis - wls$onlychild # those with no younger siblings and not
  wls$ygstchild <- case_match(wls$ygsib, 0 ~ 1, NA ~ NA, .default = 0)
  #wls %>% select(edbro, edsis, ygbro, ygsis, onlychild, edstchild, ygstchild)

## Cognitive Ability
wls$cogability <- wls$w1all3p

## Geographical Area
wls <- wls %>% mutate(geoarea = case_match(wladmarea,
  c(11,31,21) ~ 1, # BBG
```

```

c(33,22,34,35) ~ 2, # TZZM
c(23,36,37,38) ~ 3, # ZZZT
c(39,24,40,25,41) ~ 4, # YJJNN
c(12,42,43) ~ 5, # GGP
c(32,45,44,46) ~ 6)) # YHDP

## Select variables
wls <- wls %>% select(
  stud_id, wlsch_id, wlpriv, wlurban3, wlscorea, wladmarea, geoarea, # student & school info
  male, onlychild, edstchild, ygstchild, cogability) # student background & outcome

```

3 Parent Survey Data

```

## Stamp parent data
path_prt <- file.path(raw_root, "TEPS", "w1_j_p_lv7.0.dta")
file_stamp(path_prt)

```

	file	size_mb	modified	md5
1	w1_j_p_lv7.0.dta	3.7	2021-07-16	f9cd51311969f9af7155f5b2ffbae922

```

## Read family data
wlp <- read_dta(path_prt) # n = 19779

## Filter data
wlp <- wlp %>% filter(wlp101 == 1) # responded by one of the parents, n = 19053
#wlp %>% count(wlp101)

```

```

## Marriage
wlp$marriage <- case_match(wlp$wlp103, c(1,2) ~ 1, c(97,99) ~ NA, .default = 0)
#wlp %>% count(marriage)

## Ethnicity (1TW, 2MainLand, 3Indi&Other)
wlp$faethn <- case_match(wlp$w1faethn, c(1,2) ~ 1, 3 ~ 2, c(4,5) ~ 3, c(97,99) ~ 9)
wlp$moethn <- case_match(wlp$w1moethn, c(1,2) ~ 1, 3 ~ 2, c(4,5) ~ 3, c(97,99) ~ 9)
#xtabs(~faethn + moethn, wlp) %>% addmargins(FUN = sum)

```

```

wlp <- wlp %>%
  mutate(pethn = case_when(
    faethn == 1 & moethn == 1 ~ 1, # both TW
    faethn == 1 & moethn >= 3 ~ 2, # 1 TW + 1 Non-Main
    faethn >= 3 & moethn == 1 ~ 2,
    faethn == 2 & moethn == 2 ~ 3, # both MainLand
    faethn == 2 | moethn == 2 ~ 4, # 1 MainLand
    faethn == 3 & moethn == 3 ~ 5, # Other (both Indi/Other)
    .default = NA))
#wlp %>% count(pethn)

wlp <- wlp %>% mutate(faethn = case_match(faethn, 9 ~ NA, .default = faethn)) %>%
  mutate(moethn = case_match(moethn, 9 ~ NA, .default = moethn))

## Education
wlp$faedu <- case_match(wlp$w1faedu, c(6,97,99) ~ 0, .default = wlp$w1faedu)
wlp$moedu <- case_match(wlp$w1moedu, c(6,97,99) ~ 0, .default = wlp$w1moedu)
#xtabs(~ faedu + moedu, wlp) %>% addmargins(FUN = sum)

wlp <- wlp %>%
  mutate(pedu = case_when(
    faedu >= moedu ~ faedu,
    faedu < moedu ~ moedu)) %>%
  mutate(pedu = case_match(pedu, 0 ~ NA, .default = pedu))
#wlp %>% count(pedu)

wlp$pedu2 <- case_match(wlp$pedu, c(1,2,3) ~ 0, c(4,5) ~ 1, NA ~ NA)
#wlp %>% count(pedu2)

wlp <- wlp %>% mutate(faedu = case_match(faedu, 0 ~ NA, .default = faedu)) %>%
  mutate(moedu = case_match(moedu, 0 ~ NA, .default = moedu))

## Income (<2, 2-5, 5-10, 10-15, >15)
wlp$income <- case_match(wlp$w1p515, 6 ~ 5, c(97,99) ~ NA, .default = wlp$w1p515)
#wlp %>% count(income)

wlp$income2 <- case_match(wlp$income, c(1,2,3) ~ 0, c(4,5) ~ 1, NA ~ NA)
#wlp %>% count(income2)

```

```

## Parent SES (Education x Income)
wlp$pses <- case_when(
  wlp$pedu2 == 0 & wlp$income2 == 0 ~ 1,
  wlp$pedu2 == 0 & wlp$income2 == 1 ~ 2,
  wlp$pedu2 == 1 & wlp$income2 == 0 ~ 3,
  wlp$pedu2 == 1 & wlp$income2 == 1 ~ 4)

## Occupation (1Labor, 2Service, 3Public, 4Expert, 5Other)
wlp$faocc <- case_match(wlp$w1faocc, 7 ~ 1, 6 ~ 2, c(1,2) ~ 3, c(3,4,5) ~ 4, c(8,9) ~ 0, c(9
wlp$moocc <- case_match(wlp$w1moocc, 7 ~ 1, 6 ~ 2, c(1,2) ~ 3, c(3,4,5) ~ 4, c(8,9) ~ 0, c(9
  #xtabs(~ faocc + moocc, wlp) %>% addmargins(FUN = sum)

wlp <- wlp %>%
  mutate(pocc = case_when(
    faocc == 4 | moocc == 4 ~ 4, # 4: Exp
    faocc == 3 | moocc == 3 ~ 3,
    faocc == 2 | moocc == 2 ~ 2,
    faocc == 1 | moocc == 1 ~ 1,
    faocc == 0 | moocc == 0 ~ 5, # 5: Other
    faocc == -1 & moocc == -1 ~ NA))
  #wlp %>% count(pocc)

wlp <- wlp %>% mutate(faocc = case_when(
  faocc == -1 & moocc != -1 ~ 5,
  faocc == -1 & moocc == -1 ~ NA,
  faocc == 0 ~ 5,
  .default = faocc))
wlp <- wlp %>% mutate(moocc = case_when(
  moocc == -1 & faocc <= 5 ~ 5,
  moocc == -1 & is.na(faocc) ~ NA,
  moocc == 0 ~ 5,
  .default = moocc))

## Employer
wlp$faboss <- case_match(wlp$w1faself, c(3,4) ~ 1, c(97,99) ~ 9, .default = 0)
wlp$moboss <- case_match(wlp$w1moself, c(3,4) ~ 1, c(97,99) ~ 9, .default = 0)
  #xtabs(~ faboss + moboss, wlp)

wlp <- wlp %>%
  mutate(pboss = case_when(

```

```

    faboss == 1 | moboss == 1 ~ 1,
    faboss == 9 & moboss == 9 ~ NA,
    .default = 0))
#w1p %>% count(pboss)

## Management (excluding self-employers)
w1p$famgm <- case_match(w1p$w1famgm, c(1,2) ~ 1, c(97,99) ~ 9, .default = 0)
w1p$momgm <- case_match(w1p$w1momgm, c(1,2) ~ 1, c(97,99) ~ 9, .default = 0)
#xtabs(~ w1faself + famgm, w1p)

w1p <- w1p %>%
  mutate(famgm = case_when(
    w1faself == 2 | w1faself == 3 | w1faself == 4 | w1faself == 6 ~ 0,
    .default = famgm)) %>%
  mutate(momgm = case_when(
    w1moself == 2 | w1moself == 3 | w1moself == 4 | w1moself == 6 ~ 0,
    .default = momgm))
#xtabs(~ w1faself + famgm, w1p)
#xtabs(~ famgm + momgm, w1p)

w1p <- w1p %>%
  mutate(pmgm = case_when(
    famgm == 1 | momgm == 1 ~ 1,
    famgm == 9 & momgm == 9 ~ NA,
    .default = 0))
#w1p %>% count(pmgm)

w1p <- w1p %>% mutate(faboss = case_match(faboss, 9 ~ NA, .default = faboss)) %>%
  mutate(moboss = case_match(moboss, 9 ~ NA, .default = moboss)) %>%
  mutate(famgm = case_match(famgm, 9 ~ NA, .default = famgm)) %>%
  mutate(momgm = case_match(momgm, 9 ~ NA, .default = momgm))

## Educational Expectation (above college)
w1p$pexpedu <- case_match(w1p$w1p510a, c(4,5) ~ 1, c(97,99) ~ NA, .default = 0)
#w1p %>% count(pexpedu)

## Book Exhibition
w1p$book <- case_match(w1p$w1p301, c(3,4) ~ 1, c(97,99) ~ NA, .default = 0)
#w1p %>% count(bookexpo)

```

```

## Art Performance
wlp$art <- case_match(wlp$wlp302, c(3,4) ~ 1, c(97,99) ~ NA, .default = 0)
#wlp %>% count(bookexpo)

## Volunteer
wlp$volunt <- case_match(wlp$wlp303, c(3,4) ~ 1, c(97,99) ~ NA, .default = 0)
#wlp %>% count(volunt)

## Relocation (actual or on paper)
wlp$reloc <- case_match(wlp$wlp501, 2 ~ 1, c(97,99) ~ NA, .default = 0)
#wlp %>% count(reloc)

## Academic Communication
wlp$acadcomm <- case_match(wlp$wlp310, c(3,4) ~ 1, c(97,99) ~ NA, .default = 0)
#wlp %>% count(acadcomm)

## Talent Class
wlp$talentcls <- case_match(wlp$wlp304, 2 ~ 1, c(97,99) ~ NA, .default = 0)

wlp <- wlp %>% mutate(ntaxentcls =
  wlp3052 + wlp3053 + wlp3054 + wlp3055 + wlp3056) %>%
  mutate(ntaxentcls = case_match(ntaxentcls, 495 ~ 0, .default =ntaxentcls))
#wlp %>% count(ntaxentcls)

## Travel
wlp$travel <- case_match(wlp$wlp504, c(2,3) ~ 1, c(97,99) ~ NA, .default = 0)

## Abroad Study
wlp$abstudy <- case_match(wlp$wlp503, 2 ~ 1, c(97,99) ~ NA, .default = 0)

## Saving for Educaion
wlp$edusav <- case_match(wlp$wlp507, 2 ~ 1, c(97,99) ~ NA, .default = 0)
#wlp %>% count(edusav)

```

```
## Select variables
wlp <- wlp %>% select(stud_id,
  marriage, faethn, moethn, pethn, # basic info.
  faedu, moedu, pedu, pedu2, income, income2, pses, # SES
  faocc, moocc, pocc, faboss, moboss, pboss, famgm, momgm, pmgm, # occupation
  pexpedu, # parenting
  book, art, volunt, reloc, # parenting-accompany
  acadcomm, talentcls, ntalentcls, travel, abstudy, edusav) # parenting-outsource

## Merge data
teps <- inner_join(wls, wlp, by = "stud_id") # n = 20055/19053 -> 19053
```

4 School Data

```
## Stamp score data
path_scr <- file.path(raw_root, "TEPS", "w1j_score.dta")
file_stamp(path_scr)
```

	file	size_mb	modified	md5
1	w1j_score.dta	2	2022-01-18	08b531bdc94914e3f2a8b1bfa42c50ed

```
## Attach school id to score data
score <- read_dta(path_scr) %>% # n = 18929
  select(stud_id, score_fin) %>% drop_na() # n = 18032

score <- wls %>% select(stud_id, wlsch_id, w1priv, w1urban3, w1admarea, geoarea) %>%
  right_join(score, by = "stud_id") # merge with wls to get school id

## Create a list of schools with their mean scores
schscore <- score %>% group_by(wlsch_id) %>% # create school list
  summarise(n = n(), meanscore = mean(score_fin), priv = first(w1priv),
    urban = first(w1urban3), admarea = first(w1admarea), geoarea = first(geoarea)) %>%
  arrange(desc(meanscore)) %>% # sort by final score
  filter(n >= 10) # n = 332 -> 331

## Public and Private Schools by geoarea
```

```

teps %>% group_by(wlsch_id) %>%
  summarise(priv = first(wlpriv), admarea = first(wladmarea), geoarea = first(geoarea)) %>%
  xtabs(~ priv + geoarea, .)

```

```

      geoarea
priv  1  2  3  4  5  6
0  97 30 65 29 36 22
1  18  7  9 14  6  0

```

```

## Define Star Schools (top 54/279=19.35% of public schools)
schscore <- schscore %>% filter(priv == 0) %>%
  group_by(geoarea) %>%
  mutate(rank_bygeo = rank(-meanscore)) %>% # rank school by sector
  mutate(star = case_when(
    geoarea == 1 & rank_bygeo <= round(97*0.1935) ~ 1,
    geoarea == 2 & rank_bygeo <= round(30*0.1935) ~ 1,
    geoarea == 3 & rank_bygeo <= round(65*0.1935) ~ 1,
    geoarea == 4 & rank_bygeo <= round(29*0.1935) ~ 1,
    geoarea == 5 & rank_bygeo <= round(36*0.1935) ~ 1,
    geoarea == 6 & rank_bygeo <= round(22*0.1935) ~ 1,
    .default = 0)) %>%
  ungroup() %>%
  right_join(schscore)

```

Joining with `by = join\_by(wlsch\_id, n, meanscore, priv, urban, admarea, geoarea)`

```

schscore %>% xtabs(~ star + geoarea, .)

```

```

      geoarea
star  1  2  3  4  5  6
0  78 24 51 23 29 18
1  19  6 13  6  7  4

```

```

schscore$star <- schscore$star %>% replace_na(0)

```

```

## Merge data
teps <- schscore %>% select(wlsch_id, meanscore, star) %>%

```

```
right_join(teps, by = "wlsch_id") %>% # n = 19053 -> 19053
left_join(score) %>% # merge final scores
arrange(stud_id)
```

Joining with `by = join\_by(wlsch\_id, stud\_id, wlpriv, wlurban3, wladmarea, geoarea)`

```
#teps %>% filter(is.na(star)) %>% distinct(wlsch_id) # two schools with final scores miss

teps$star <- tepts$star %>% replace_na(0)
#teps %>% filter(star == 1) %>% distinct(wlsch_id)

## Create Schooltype
teps$schtype <- case_when(
  tepts$wlpriv == 1 & tepts$star == 0 ~ 2,
  tepts$wlpriv == 0 & tepts$star == 1 ~ 3,
  .default = 1)
#teps %>% count(schtype)
```

5 Write

```
## Check missing data
#sapply(teps, function(y) sum(is.na(y))) %>% data.frame(na_count = .)

## Rename and order columns
teps <- tepts %>% rename(
  sch_id = wlsch_id,
  priv = wlpriv,
  urban = wlurban3,
  scarea = wiscarea,
  admarea = wladmarea,
  finscore = score_fin)

teps <- tepts %>% select(stud_id,
  sch_id, urban, scarea, admarea, geoarea, meanscore, priv, star, schtype, # school
  male, cogability, finscore, # student
  onlychild, edstchild, ygstchild, marriage, faethn, moethn, pethn, # family
```

```

faedu, moedu, pedu, pedu2, income, income2, pses, # parental SES
faocc, moocc, pocc, faboss, moboss, pboss, famgm, momgm, pmgm, # parental occupation
pexpedu, # parenting
book, art, volunt, reloc, # parenting-accompany
acadcomm, talentcls, ntalentcls, travel, abstudy, edusav) # parenting-outsource

## Export data
write_dta(temps, "data/TEPS-Cleaned.dta")
pin_dta_timestamp("data/TEPS-Cleaned.dta")
file_stamp("data/TEPS-Cleaned.dta")

```

	file	size_mb	modified	md5
1	TEPS-Cleaned.dta	6.7	2026-10-01	fb98e0d36e2c5e5616b339120303ca8a

End of replication document

```
sessionInfo()
```

```

R version 4.5.3 (2026-03-11)
Platform: aarch64-apple-darwin20
Running under: macOS Tahoe 26.5.2

```

```
Matrix products: default
```

```

BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;

```

```
locale:
```

```
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
```

```
time zone: America/New_York
```

```
tzcode source: internal
```

```
attached base packages:
```

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

```
other attached packages:
```

```
[1] haven_2.5.5    lubridate_1.9.5 forcats_1.0.1  stringr_1.6.0
```

```
[5] dplyr_1.2.1      purrr_1.2.2      readr_2.2.0      tidyr_1.3.2
[9] tibble_3.3.1     ggplot2_4.0.3    tidyverse_2.0.0
```

loaded via a namespace (and not attached):

```
[1] gtable_0.3.6      jsonlite_2.0.0    compiler_4.5.3     tidyselect_1.2.1
[5] scales_1.4.0      yaml_2.3.12       fastmap_1.2.0      R6_2.6.1
[9] generics_0.1.4    knitr_1.52        pillar_1.11.1      RColorBrewer_1.1-
3
[13] tzdb_0.5.0        rlang_1.3.0       stringi_1.8.9      xfun_0.61
[17] S7_0.2.2          otl_0.2.0         timechange_0.4.0    cli_3.6.6
[21] withr_3.0.3       magrittr_2.0.5    digest_0.6.39      grid_4.5.3
[25] rstudioapi_0.19.0 hms_1.1.4         lifecycle_1.0.5    vctrs_0.7.3
[29] evaluate_1.0.5    glue_1.8.1        farver_2.1.2       rmarkdown_2.32
[33] tools_4.5.3       pkgconfig_2.0.3   htmltools_0.5.9
```